

```
!wget -q https://go.dev/dl/go1.21.5.linux-amd64.tar.gz
!tar -C /usr/local -xzf go1.21.5.linux-amd64.tar.gz
import os
os.environ['PATH'] += ':/usr/local/go/bin'
!go version
```

```
go version go1.21.5 linux/amd64
```

```
!git clone https://github.com/INMAP-PAVITRA/InMAP-PAVITRA.git
%cd InMAP-PAVITRA
```

```
Cloning into 'InMAP-PAVITRA'...
remote: Enumerating objects: 16564, done.
remote: Counting objects: 100% (13/13), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 16564 (delta 4), reused 11 (delta 2), pack-reused 16551 (from 1)
Receiving objects: 100% (16564/16564), 100.22 MiB | 20.38 MiB/s, done.
Resolving deltas: 100% (3032/3032), done.
Updating files: 100% (18305/18305), done.
/content/InMAP-PAVITRA
```

```
!GO111MODULE=on go build ./cmd/inmap
```

```
go: downloading github.com/cenkalti/backoff v2.2.1+incompatible
go: downloading github.com/ctessum/geom v0.2.10
go: downloading github.com/ctessum/gobra v0.0.0-20180516235632-ddfa5eeb3017
go: downloading github.com/lnashier/viper v0.0.0-20180730210402-cc7336125d12
go: downloading github.com/skratchdot/open-golang v0.0.0-20160302144031-75fb7ed4208c
go: downloading github.com/spf13/cast v1.2.0
go: downloading github.com/spf13/cobra v0.0.3
go: downloading github.com/spf13/pflag v1.0.5
go: downloading gocloud.dev v0.23.0
go: downloading google.golang.org/grpc v1.37.0
go: downloading github.com/ctessum/polyclip-go v1.0.2-0.20200417141046-48e92ea36ddd
go: downloading gonum.org/v1/gonum v0.0.0-20191009222026-5d5638e6749a
go: downloading golang.org/x/net v0.7.0
go: downloading github.com/Knetic/govaluate v3.0.1-0.20171022003610-9aa49832a739+incompatible
go: downloading github.com/ctessum/atmos v0.0.0-20170526022537-cba69f7ca647
go: downloading github.com/ctessum/cdf v0.0.0-20181201011353-edced208ea9d
go: downloading github.com/ctessum/sparse v0.0.0-20181201011727-57d6234a2c9d
go: downloading github.com/ctessum/unit v0.0.0-20160621200450-755774ac2fcb
go: downloading github.com/jonas-p/go-shp v0.1.2-0.20190401125246-9fd306ae10a6
go: downloading github.com/aws/aws-sdk-go v1.38.35
go: downloading github.com/improbable-eng/grpc-web v0.0.0-20190113155728-0c7a81a25d11
go: downloading k8s.io/api v0.20.1
go: downloading k8s.io/apimachinery v0.20.1
go: downloading k8s.io/client-go v0.20.1
go: downloading github.com/golang/protobuf v1.5.2
go: downloading github.com/cenkalti/backoff/v4 v4.1.1
go: downloading github.com/ctessum/requestcache/v2 v2.0.0
go: downloading github.com/ctessum/requestcache v1.0.1
go: downloading github.com/ctessum/requestcache/v4 v4.0.0
go: downloading github.com/jackc/pgx/v4 v4.12.0
go: downloading github.com/mattn/go-sqlite3 v1.11.0
go: downloading github.com/tealeg/xlsx v1.0.3
go: downloading github.com/gonum/floats v0.0.0-20181209220543-c233463c7e82
go: downloading github.com/fsnotify/fsnotify v1.4.9
go: downloading github.com/hashicorp/hcl v0.0.0-20171017181929-23c074d0ecec
go: downloading github.com/magiconair/properties v1.7.3
go: downloading github.com/mitchellh/mapstructure v1.4.1
go: downloading github.com/pelletier/go-toml v1.0.1
go: downloading github.com/spf13/afero v1.2.2
go: downloading github.com/spf13/jwalterweatherman v0.0.0-20170901151539-12bd96e66386
go: downloading gopkg.in/yaml.v2 v2.4.0
```

```

go: downloading go.opencensus.io v0.23.0
go: downloading cloud.google.com/go v0.81.0
go: downloading cloud.google.com/go/storage v1.15.0
go: downloading github.com/google/wire v0.5.0
go: downloading github.com/googleapis/gax-go/v2 v2.0.5
go: downloading golang.org/x/oauth2 v0.0.0-20210427180440-81ed05c6b58c
go: downloading github.com/googleapis/gax-go v2.0.0+incompatible
go: downloading google.golang.org/api v0.46.0
go: downloading google.golang.org/genproto v0.0.0-20210506142907-4a47615972c2
go: downloading github.com/gorilla/websocket v1.4.2
go: downloading github.com/rs/cors v1.3.0
go: downloading github.com/gogo/protobuf v1.3.2
go: downloading gopkg.in/inf.v0 v0.9.1
go: downloading github.com/google/gofuzz v1.1.0
go: downloading k8s.io/klog/v2 v2.4.0
go: downloading sigs.k8s.io/structured-merge-diff/v4 v4.0.2

```

```
!wget -q --show-progress "https://zenodo.org/records/3403934/files/evaldata_v1.6.1.zip?download=1" -O
```

```
/content/evaldata_v 100%[=====>] 2.02G 2.21MB/s in 15m 16s
```

```
!unzip -q /content/evaldata_v1.6.1.zip -d /content/
!ls -la /content/evaldata_v1.6.1
```

```

total 2276556
drwxr-xr-x 6 root root      4096 Sep 10  2019 .
drwxr-xr-x 1 root root      4096 Jun 16 09:02 ..
drwxrwxr-x 2 root root      4096 Aug 22  2016 2005_emissions
-rw-r--r-- 1 root root 49368451 Feb 27  2014 annual_all_2005.csv
-rw-rw-r-- 1 root root 22662576 Aug 25  2017 census2013blckgrp.dbf
-rw-rw-r-- 1 root root   411 Aug 24  2017 census2013blckgrp.prj
-rw-rw-r-- 1 root root   2779 Aug 25  2017 census2013blckgrp_README.txt
-rw-rw-r-- 1 root root 1119525316 Aug 24  2017 census2013blckgrp.shp
-rw-rw-r-- 1 root root  1726748 Aug 24  2017 census2013blckgrp.shx
drwxr-x--- 4 root root      4096 Nov 15  2016 FuelScenarios
-rw-r----- 1 root root 727596732 Jun 28  2016 InMAPData_v1.2.0.ncf
-rw-r--r-- 1 root root 218190478 Sep 10  2019 InMAPData_v1.6.1.gob
drwxrwxr-x 2 root root      4096 Aug 22  2016 la_test
-rw-rw-r-- 1 root root 482152 Aug 25  2017 mortalityRates2013.dbf
-rw-rw-r-- 1 root root   411 Aug 25  2017 mortalityRates2013.prj
-rw-rw-r-- 1 root root   1726 Aug 25  2017 mortalityRates2013_README.txt
-rw-rw-r-- 1 root root 190692668 Aug 25  2017 mortalityRates2013.shp
-rw-rw-r-- 1 root root  24972 Aug 25  2017 mortalityRates2013.shx
-rw-rw-r-- 1 root root   3692 Aug 25  2017 README.txt
drwxrwxr-x 2 root root      4096 Nov 15  2016 singleSource
-rw-r--r-- 1 root root  10177 Feb 13  2014 states.dbf
-rw-r--r-- 1 root root    297 Feb 13  2014 states.prj
-rw-r--r-- 1 root root  830916 Feb 13  2014 states.shp
-rw-r--r-- 1 root root    516 Feb 13  2014 states.shx

```

```

from google.colab import files
uploaded = files.upload()

```

[Show hidden output](#)

```

!pip install netCDF4 -q --break-system-packages 2>/dev/null
import netCDF4 as nc
ds = nc.Dataset('/content/evaldata_v1.6.1/la_test/inmapData.ncf')
print(ds)
print()
for v in ds.variables:
    print(v, ds.variables[v].shape)

```

```
10.1/10.1 MB 70.2 MB/s eta 0:00:00
```

```
1.6/1.6 MB 93.5 MB/s eta 0:00:00
```

```

<class 'netCDF4.Dataset'>
root group (NETCDF3_CLASSIC data model, file format NETCDF3):
  comment: InMAP meteorology and baseline chemistry data file
  x0: -2004000.0
  y0: -540000.0
  dx: 12000.0
  dy: 12000.0
  nx: 24
  ny: 20
  data_version: 1.2.0
  dimensions(sizes): x(24), y(20), z(27), xStagger(25), yStagger(21), zStagger(28)
  variables(dimensions): float32 aVOC(z, y, x), float32 gNH(z, y, x), float32 ParticleDryDep(z, y,
groups:

aVOC (27, 20, 24)
gNH (27, 20, 24)
ParticleDryDep (27, 20, 24)
SO2DryDep (27, 20, 24)
M2d (27, 20, 24)
UAvg (27, 20, 25)
VDeviation (27, 21, 24)
aOrgPartitioning (27, 20, 24)
Temperature (27, 20, 24)
Sclass (27, 20, 24)
alt (27, 20, 24)
NHPartitioning (27, 20, 24)
Kxxyy (27, 20, 24)
Kzz (28, 20, 24)
WAvg (28, 20, 24)
bOrgPartitioning (27, 20, 24)
SPartitioning (27, 20, 24)
VAvg (27, 21, 24)
aSOA (27, 20, 24)
Dz (27, 20, 24)
NH3DryDep (27, 20, 24)
WindSpeed (27, 20, 24)
S1 (27, 20, 24)
UDeviation (27, 20, 25)
gNO (27, 20, 24)
pNH (27, 20, 24)
TotalPM25 (27, 20, 24)
bVOC (27, 20, 24)
OtherGasWetDep (27, 20, 24)
WindSpeedInverse (27, 20, 24)
WindSpeedMinusThird (27, 20, 24)
WindSpeedMinusOnePointFour (27, 20, 24)
SO2oxidation (27, 20, 24)
LayerHeights (28, 20, 24)
M2u (27, 20, 24)
pS (27, 20, 24)
NOxDryDep (27, 20, 24)
ParticleWetDep (27, 20, 24)
NOPartitioning (27, 20, 24)
pNO (27, 20, 24)
gS (27, 20, 24)

```

```

config_content = '''InMAPData = "/content/evaldata_v1.6.1/la_test/inmapData.ncf"
VariableGridData = "/content/evaldata_v1.6.1/la_test/inmapVarGrid.gob"

EmissionsShapefiles = [
    "/content/evaldata_v1.6.1/la_test/testEmis.shp"
]

EmissionUnits = "tons/year"
HTTPAddress = ":8080"

OutputFile = "/content/InMAP-PAVITRA/laTest/laTest_output.shp"
LogFile = "/content/InMAP-PAVITRA/laTest/logfile.log"

```

```

[OutputVariables]
TotalPM25 = "PrimaryPM25 + pNH4 + pSO4 + pNO3 + SOA"
TotalPopD = "(exp(log(1.078)/10 * TotalPM25) - 1) * TotalPop * AllCause / 100000"
TotalPop = "TotalPop"
PrimPM25 = "PrimaryPM25"
PSO4 = "pSO4"
PNO3 = "pNO3"
PNH4 = "pNH4"
SOA = "SOA"
SOx = "SOx"
NH3 = "NH3"
NOx = "NOx"
BasePM25 = "BaselineTotalPM25"
WindSpeed = "WindSpeed"

[VarGrid]
VariableGridXo = -2004000.0
VariableGridYo = -540000.0
VariableGridDx = 12000.0
VariableGridDy = 12000.0
Xnests = [24]
Ynests = [20]
HiResLayers = 1
GridProj = "+proj=lcc +lat_1=33.000000 +lat_2=45.000000 +lat_0=40.000000 +lon_0=-97.000000 +x_0=0 +y_0=0"
PopDensityThreshold = 0.0055
PopThreshold = 40000.0
PopConcThreshold = 0.000000001

CensusFile = "/content/evaldata_v1.6.1/la_test/census2012.shp"
CensusPopColumns = ["TotalPop"]
PopGridColumn = "TotalPop"

MortalityRateFile = "/content/evaldata_v1.6.1/la_test/MortalityRate.shp"

[VarGrid.MortalityRateColumns]
AllCause = "TotalPop"
...

with open('/content/laTestConfig.toml', 'w') as f:
    f.write(config_content)

print("Config updated")

```

Config updated

```

!mkdir -p /content/InMAP-PAVITRA/laTest
%cd /content/InMAP-PAVITRA
!./inmap run steady --config=/content/laTestConfig.toml

```

```

SOA pop-wtd:      0.0014%
PrimaryPM25:      0.039%
PrimaryPM25 pop-wtd: 0.0018%
NH3:              0.097%
NH3 pop-wtd:      0.0039%
pNH4:             0.036%
pNH4 pop-wtd:     0.0017%
SOx:              0.044%
SOx pop-wtd:      0.0023%
pSO4:             0.12%
pSO4 pop-wtd:     0.03%
NOx:              0.063%
NOx pop-wtd:      0.0037%
pNO3:             0.14%
pNO3 pop-wtd:     0.0063%
2026/06/16 09:55:28 iteration 8555 walltime=0.0677h Δwalltime=0.03s timestep=20s day=2.02
2026/06/16 09:55:30 iteration 8639 walltime=0.0682h Δwalltime=0.024s timestep=20s day=2.04
2026/06/16 09:55:32 iteration 8725 walltime=0.0688h Δwalltime=0.023s timestep=20s day=2.06
2026/06/16 09:55:34 iteration 8769 walltime=0.0694h Δwalltime=0.046s timestep=20s day=2.07
2026/06/16 09:55:36 iteration 8812 walltime=0.0699h Δwalltime=0.051s timestep=20s day=2.08
2026/06/16 09:55:38 iteration 8891 walltime=0.0705h Δwalltime=0.02s timestep=20s day=2.1
2026/06/16 09:55:40 iteration 8980 walltime= 0.071h Δwalltime=0.022s timestep=20s day=2.12
2026/06/16 09:55:41 Adding grid cells...
2026/06/16 09:55:41 Added 0 grid cells; there are now 12960 cells total
2026/06/16 09:55:41 Percent change since last convergence check:
VOC:              0.014%
VOC pop-wtd:      0.00062%
SOA:              0.016%
SOA pop-wtd:      0.00078%
PrimaryPM25:      0.023%
PrimaryPM25 pop-wtd: 0.001%
NH3:              0.057%
NH3 pop-wtd:      0.0023%
pNH4:             0.021%
pNH4 pop-wtd:     0.00098%
SOx:              0.026%
SOx pop-wtd:      0.0013%
pSO4:             0.07%
pSO4 pop-wtd:     0.018%
NOx:              0.038%
NOx pop-wtd:      0.0022%
pNO3:             0.085%
pNO3 pop-wtd:     0.0038%
2026/06/16 09:55:41 Elapsed time: 0.071989 hours

```

```
!ls -la /content/InMAP-PAVITRA/laTest/
```

```

total 220
drwxr-xr-x  2 root root   4096 Jun 16 09:55 .
drwxr-xr-x 17 root root   4096 Jun 16 09:48 ..
-rw-r--r--  1 root root 113249 Jun 16 09:55 laTest_output.dbf
-rw-r--r--  1 root root    431 Jun 16 09:55 laTest_output.prj
-rw-r--r--  1 root root  65380 Jun 16 09:55 laTest_output.shp
-rw-r--r--  1 root root   3940 Jun 16 09:55 laTest_output.shx
-rw-r--r--  1 root root   21025 Jun 16 09:55 logfile.log

```

```

!pip install geopandas -q --break-system-packages 2>/dev/null
import geopandas as gpd

gdf = gpd.read_file('/content/InMAP-PAVITRA/laTest/laTest_output.shp')
print("Columns:", gdf.columns.tolist())
print()
print("Number of grid cells:", len(gdf))
print()
print(gdf[['TotalPM25', 'TotalPop', 'TotalPopD']].describe())

```

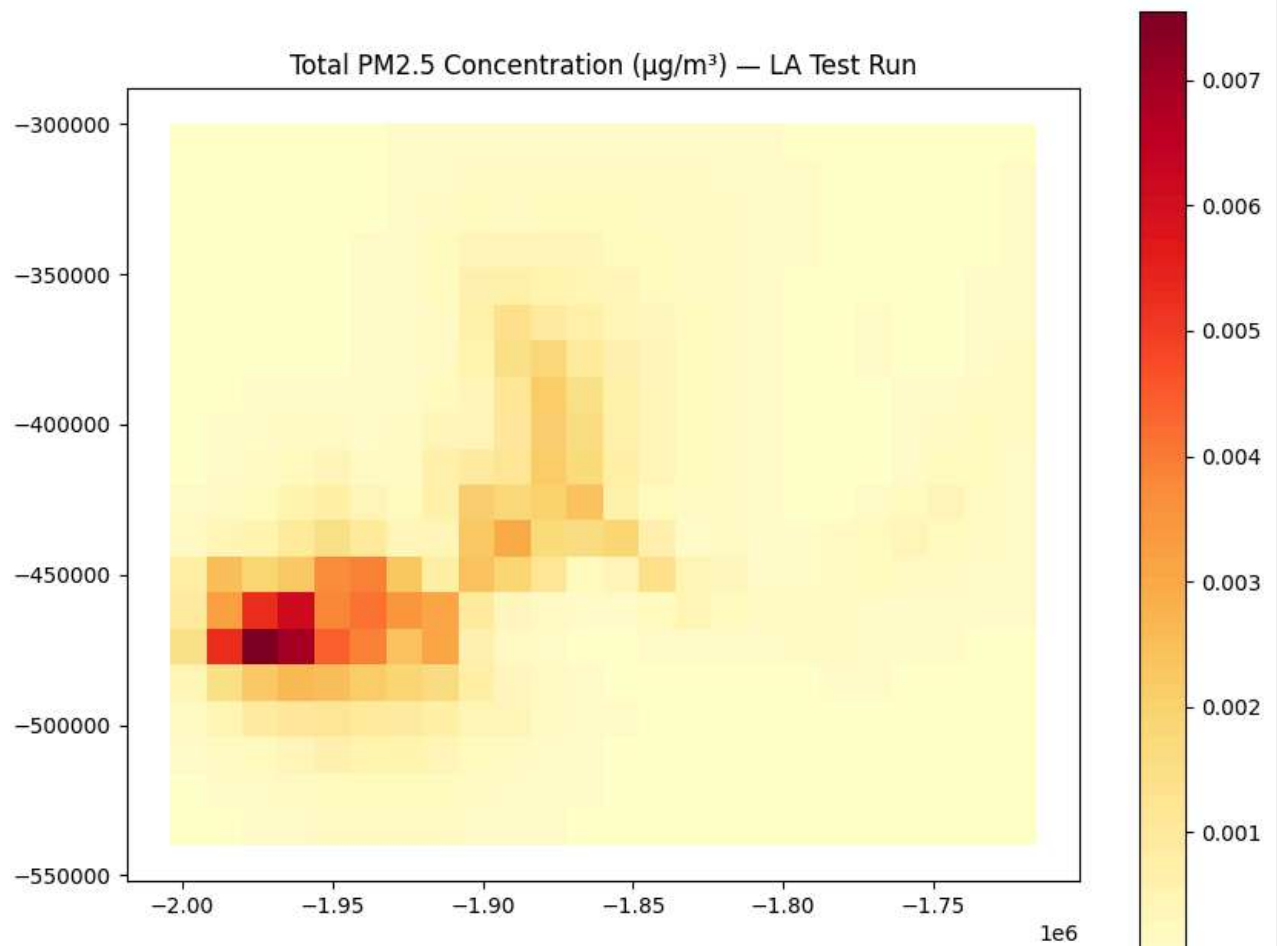
```
Columns: ['BasePM25', 'NH3', 'NOx', 'PNH4', 'PNO3', 'PSO4', 'PrimPM25', 'SOA', 'SOx', 'TotalPM25', 'Tot
```

Number of grid cells: 480

	TotalPM25	TotalPop	TotalPopD
count	4.800000e+02	480.000000	4.800000e+02
mean	4.385687e-04	35262.373688	1.934549e-03
std	9.414985e-04	102987.107845	9.009524e-03
min	2.880303e-08	0.000000	0.000000e+00
25%	1.551790e-05	0.000000	0.000000e+00
50%	8.995863e-05	44.446306	2.232373e-08
75%	3.296119e-04	4995.330813	3.036618e-05
max	7.546752e-03	853304.286139	8.527945e-02

```
import matplotlib.pyplot as plt
```

```
fig, ax = plt.subplots(figsize=(10, 8))
gdf.plot(column='TotalPM25', cmap='YlOrRd', legend=True, ax=ax)
ax.set_title('Total PM2.5 Concentration ( $\mu\text{g}/\text{m}^3$ ) - LA Test Run')
plt.show()
```



```
# CHECK STATUS
!cat /content/inmap_run.log
!ps aux | grep inmap
```

[Show hidden output](#)

```
!tail -20 /content/inmap_run.log
!ps aux | grep inmap
!free -h
```

Show hidden output

```
!ls -la /content/evaldata_v1.6.1/la_test
```

Show hidden output

```
!pkill -9 inmap
```